SUBMISSION HIGHLIGHTS

Project AMBROSIUM v10.0 • Lab on a Chip

• 10 MHz 14-bit ADC & Artix-7 FPGA FSM guard (y_prev < 16'sd2000) for metrology.

• Electrodialysis expands Debye length 0.80 nm to 4.98 nm under V_gate < 1.23 V.

• 10,000-channel SFAT MEMS array achieves >= -28.3 dB isolation at 25 um pitch.

• WR-340 2.45 GHz RF flash gelation in tau = 0.85 s with aerogel CTE buffering.

• 1D thermal PDE finite difference solver proves Courant ratio r = 0.0572 << 0.50.